

Day 3: Introduction to IP Addressing

Task 1 — IPv4 vs IPv6

IPv4	IPv6
IPv4 uses 32-bit addresses.	IPv6 uses 128-bit addresses.
Example: 192.168.1.10	Example: 2001:db8::10
It has a smaller address space.	It provides a much larger address space.
IPv4 addresses are written in decimal format.	IPv6 addresses are written in hexadecimal format.
Example mask: 255.255.255.0	Uses prefix length such as /64.

Short Explanation:

IPv4 is the older and widely used IP addressing system. IPv6 was developed to provide a much larger number of IP addresses and support the growing number of Internet-connected devices.

Task 2 — Static IPv4 Address Assignment

The home network is **192.168.10.0/24**.

For a /24 network, the subnet mask is
255.255.255.0.

I assigned different static IP addresses to each device.

Device	IP Address	Subnet Mask	Default Gateway
Laptop	192.168.10.10	255.255.255.0	192.168.10.1
Phone	192.168.10.11	255.255.255.0	192.168.10.1
Smart TV	192.168.10.12	255.255.255.0	192.168.10.1
Printer	192.168.10.13	255.255.255.0	192.168.10.1

All devices have unique IP addresses in the same network, so they can communicate with each other through the home router.

Task 3 — /27 Subnetting

Network: **192.168.1.0/24**

New subnet mask: **/27**

A /27 subnet has:

- **32 total addresses per subnet**
- **30 usable host addresses**
- **8 valid subnets** from a /24 network
- Block size = **32**

The first three subnet ranges are:

Subnet	Network Address	Usable Host IPs	Broadcast Address
1	192.168.1.0	192.168.1.1 – 192.168.1.30	192.168.1.31
2	192.168.1.32	192.168.1.33 – 192.168.1.62	192.168.1.63
3	192.168.1.64	192.168.1.65 – 192.168.1.94	192.168.1.95

Calculation:

$/27 \rightarrow 32 - 27 = 5$ host bits

Total addresses:

$$2^5 = 32$$

Usable hosts:

$$32 - 2 = 30$$

Number of /27 subnets inside a /24:

$$2^3 = 8$$

Task 4 — Find Network and Broadcast Address

Given:

IP Address: 10.0.5.17

Subnet Mask: 255.255.255.0

The subnet mask 255.255.255.0 is equal to /24.

Network Address

The first three octets represent the network:

Network Address: 10.0.5.0

Broadcast Address

The last address of this /24 network is:

Broadcast Address: 10.0.5.255

Valid Host Range

The first usable address is:

10.0.5.1

The last usable address is:

10.0.5.254

So the complete answer is:

Item	Address
IP Address	10.0.5.17
CIDR	/24
Network Address	10.0.5.0
First Host	10.0.5.1
Last Host	10.0.5.254
Broadcast Address	10.0.5.255

Task 5 — Subnet Calculator Verification

I used an online subnet calculator to verify the answer from Task 4.

Input:

10.0.5.17 / 255.255.255.0

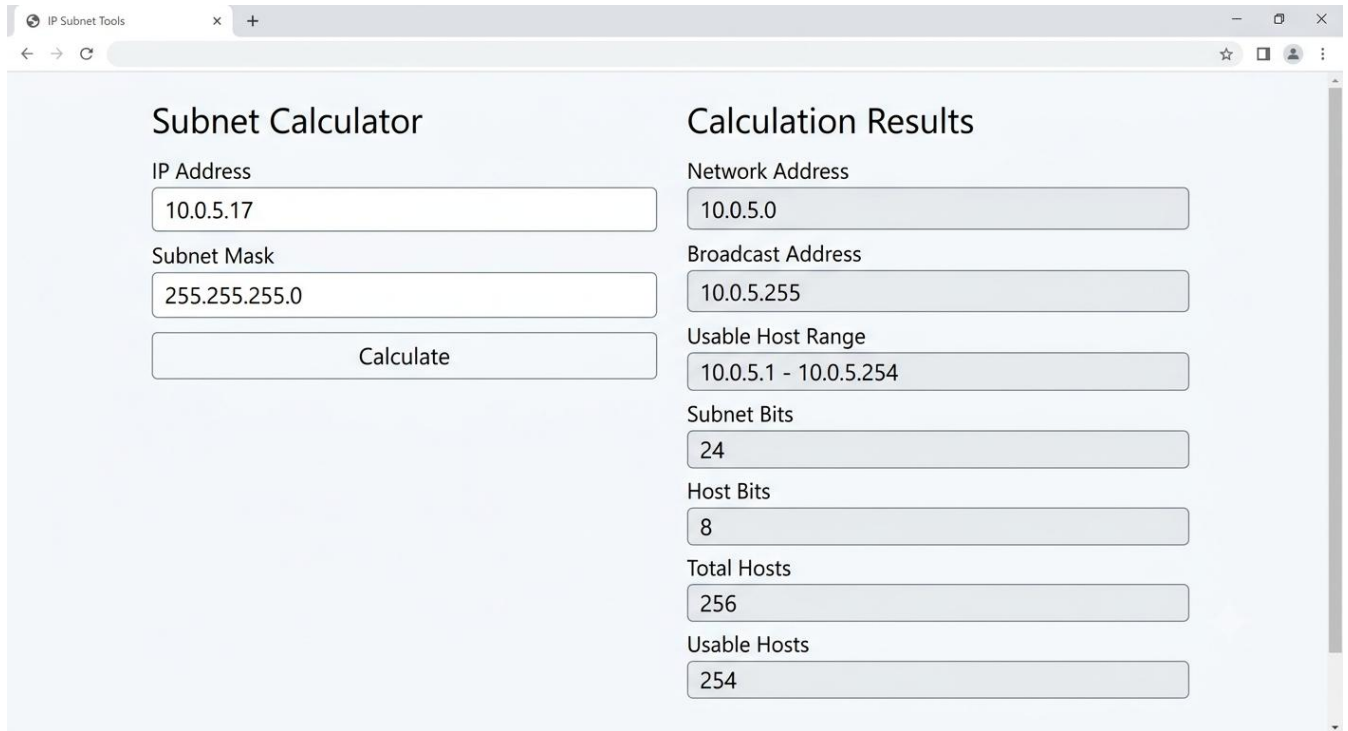
The calculator showed:

- Network: 10.0.5.0

- Broadcast: 10.0.5.255
- Usable range: 10.0.5.1 – 10.0.5.254

Result:

My manual calculation matched the output from the subnet calculator.



The screenshot shows a web browser window titled "IP Subnet Tools". The page is divided into two main sections: "Subnet Calculator" on the left and "Calculation Results" on the right. In the "Subnet Calculator" section, the "IP Address" field contains "10.0.5.17" and the "Subnet Mask" field contains "255.255.255.0". Below these fields is a "Calculate" button. The "Calculation Results" section displays the following values: "Network Address" is 10.0.5.0, "Broadcast Address" is 10.0.5.255, "Usable Host Range" is 10.0.5.1 - 10.0.5.254, "Subnet Bits" is 24, "Host Bits" is 8, "Total Hosts" is 256, and "Usable Hosts" is 254.

Subnet Calculator	Calculation Results
IP Address 10.0.5.17	Network Address 10.0.5.0
Subnet Mask 255.255.255.0	Broadcast Address 10.0.5.255
Calculate	Usable Host Range 10.0.5.1 - 10.0.5.254
	Subnet Bits 24
	Host Bits 8
	Total Hosts 256
	Usable Hosts 254

Conclusion

In this task, I learned the difference between IPv4 and IPv6, assigned static IP addresses to devices, and practiced subnetting using CIDR notation. I also calculated network, broadcast and usable host addresses and verified the result using a subnet calculator.